

Unit 03: Mathematical Operations

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Objectives

The key concepts explained in this unit, including area a learner is expected to learn and master after going through the unit are: -

- Understand what is BODMAS Rule.
- Understand conversion of symbols into signs.

Introduction

What is BODMAS Rule?

BODMAS is an acronym used for Brackets, Order, Division, Multiplication, Addition and Subtraction. In some regions, people/students use PEMDAS (Parentheses, Exponents, Multiplication, Division, Addition and Subtraction) which is the synonym or equivalent of the BODMAS rule and can be used interchangeably.

It explains the mathematical operations to be performed while solving a mathematical expression. According to this rule, if multiple brackets (vinculum, +, ×, ÷) are there in the expression, start solving inside the vinculum or bar or line bracket first, round bracket then followed by curly bracket, then square bracket and then solve the order (means power and roots etc), then division, multiplication, addition and then subtraction.

So, the BODMAS rule is used to evaluate mathematical expressions and to deal with complex calculations in a much easier way and correctly.

3.1 Definition of BODMAS Rule

According to the rule, to solve any mathematical expression, first, solve the terms written inside the brackets, and then simplify exponential terms and move ahead to division and multiplication operations, then, at last, addition and subtraction.

Here, multiplication and division can be considered as level one operations as they must be solved first, addition and subtraction can be considered as level two operations. Simplification of terms inside the brackets can be done directly. This means we can perform the operations inside the bracket in order of division, multiplication, addition, and subtraction.

Analytical Skills-I

Sticking to this order of operations in the BODMAS rule, always gives the correct answer. If there are multiple brackets in an expression, all the same types of brackets can be solved simultaneously.



Example:- $(31+2) \div (13-2) = 33 \div 11 = 3$

Look at the below diagram to understand the terms and operations denoted by the BODMAS acronym in the proper order.

There are few conditions and rules for general simplification as given below:

Condition	Rule
$p+(q+r) \Rightarrow p+q+rp+(q+r) \Rightarrow p+q+r$	Open the bracket and add the terms.
$p-(q+r) \Rightarrow p-q-rp-(q+r) \Rightarrow p-q-r$	Open the bracket and multiply the negative sign with each term inside the bracket. (All positive terms will be negative and negative terms will be positive)
$p(q+r) \Rightarrow pq+prp(q+r) \Rightarrow pq+pr$	Multiply the outside term with each term inside the round bracket

BODMAS is an acronym used to remember the order of operations to be followed during solving mathematical expressions. According to this rule, first solve the expression inside the brackets (vinculum, {}, [], ()) then solve order or of (power or roots), then division or multiplication (as division and multiplication is having the same precedence, perform whatever comes first from left to right), then solve the addition or subtraction (as addition and subtraction are having the same precedence, perform whatever comes first from left to right). It makes simplification easy and error-free.

BODMAS (Bracket's order or of Division Multiplication Addition Subtraction) rule is correct. But in some regions, people also use PEMDAS (Parentheses Exponent Multiplication Division Addition Subtraction) or BIDMAS (Brackets Indices Division Multiplication Addition Subtraction). By the way, all three acronyms are correct.

3.2 Simple Arithmetic Operation

It is a common need to simplify the expressions

formulated according to the statements of the problems related to practical life. To do this, it is essential to follow in sequence the mathematical operations given

by the term, 'BODMAS'

3.3 BODMAS

Each letter of the word BODMAS stands as follows:

B for Bracket : {, (,), }

There are four brackets, namely, - i.e., bar, (), { } and

[]. They are removed, strictly in the order -, (), { } and [].

O for Of : of

D for Division : $\div$

M for Multiplication : $\times$

A for Addition : +

S for Subtraction : -

The order of various operations in exercises involving brackets and fractions must be strictly performed according to the order of the letters of the word BODMAS.

Illustration 1: simplify

$$8\frac{1}{2} - \left[3\frac{1}{5} + 4\frac{1}{2} \text{ of } 5\frac{1}{3} \left\{ 11 - \left(3 - 1\frac{1}{4} - \frac{5}{8} \right) \right\} \right]$$

Solution : - given expression

$$= \frac{17}{2} - \left[\frac{16}{5} \div \frac{9}{2} \text{ of } \frac{16}{3} + \left\{ 11 - \left(3 - 1\frac{1}{4} - \frac{5}{8} \right) \right\} \right]$$

$$= \frac{17}{2} - \left[\frac{16}{5} \div \frac{9}{2} \text{ of } \frac{16}{3} + \frac{69}{8} \right]$$

$$= \frac{17}{2} - \left[\frac{16}{5} \div \frac{9}{2} \times \frac{16}{3} + \frac{69}{8} \right]$$

$$= \frac{17}{2} - \left[\frac{16}{5} \div \frac{24}{1} + \frac{69}{8} \right]$$

$$= \frac{17}{2} - \left[\frac{16}{5} \times \frac{1}{24} + \frac{69}{8} \right] = \frac{17}{2} - \left[\frac{16}{120} + \frac{69}{8} \right]$$

$$= \frac{17}{2} - \left[\frac{16+1035}{120} \right]$$

$$= \frac{17}{2} - \frac{1051}{120}$$

$$= \frac{1020 - 1051}{120}$$

$$= -\frac{31}{120}$$

Illustration 2: simplify

$$5\frac{1}{3} - \left\{ 4\frac{1}{3} - \left(3\frac{1}{3} - 2\frac{1}{3} - \frac{1}{3} \right) \right\}$$

Solution : given expression

$$= \frac{16}{3} - \left\{ \frac{13}{3} - \left(\frac{10}{3} - \frac{7}{3} - \frac{1}{3} \right) \right\}$$

$$= \frac{16}{3} - \left\{ \frac{13}{3} - \left(\frac{10}{3} - \frac{6}{3} \right) \right\}$$

$$= \frac{16}{3} - \left\{ \frac{13}{3} - \frac{4}{3} \right\}$$

$$= \frac{16}{3} - \left(\frac{9}{3} \right) = \frac{16}{3} - \frac{9}{3}$$

$$= \frac{7}{3}$$

$$= 2\frac{1}{3}$$

3.4 Use of Algebraic Formulae

The following formulae are sometimes found useful in dealing with the simplification :

$$1. (a+b)^2 = a^2 + 2ab + b^2$$

$$2. (a-b)^2 = a^2 - 2ab + b^2$$

$$3. (a+b)^2 + (1-b)^2 = 2(a^2 + b^2)$$

$$4. (a+b)^2 - (a-b)^2 = 4ab$$

$$5. (a^2 - b^2) = (a+b)(a-b)$$

$$6. (a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3 = a^3 + b^3 + 3ab(a+b)$$

$$7. (a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$$

$$= a^3 - b^3 - 3ab(a-b)$$

Analytical Skills-I

$$8. a^3 + b^3 = (a+b)(a^2-ab+b^2)$$

$$9. a^3 - b^3 = (a-b)(a^2+ab+b^2)$$

$$10. \frac{a^3+b^3+c^3-3abc}{a^2+b^2+c^2-ab-bc-ca} = (a+b+c)$$

$$11. a^4 - a^4 = (a^2 + b^2)(a+b)(a-b)$$

Illustration 3: simplify the following

$$1) \quad 0.32 \times 0.32 + 0.64 \times 0.68 \times 0.68 \times 0.68$$

Solution:- Given expression = $0.32 \times 0.32 + 2 \times 0.32 \times 0.68 + 0.68 \times 0.68$

$$= (0.32)^2 + 2 \times 0.32 \times 0.68 + (0.68)^2$$

$$= (0.32 + 0.68)^2 \quad [a^2 + 2ab + b^2 = (a + b)^2]$$

$$= 1^2$$

$$= 1$$

$$ii) 2.45 \times 2.45 - 0.9 \times 2.45 + 0.45 \times 0.45$$

Solution: Given expression = $2.45 \times 2.45 - 2 \times 2.45 \times 0.45 + 0.45 \times 0.45$

$$= (2.45)^2 - 2 \times 2.45 \times 0.45 + (0.45)^2$$

$$= (2.45 - 0.45)^2 \quad [a^2 - 2ab + b^2 = (a - b)^2]$$

$$= (2)^2$$

$$= 4$$

$$iii) \frac{7 \times \{(146+92)^2 - (146-92)^2\}}{(146)^2 + (92)^2}$$

Solution: given expression

$$= \frac{7 \times 2\{(146)^2 + (92)^2\}}{(146)^2 + (92)^2} \quad [\because (a+b)^2 + (a-b)^2 = 2(a^2 + b^2)]$$

$$= 14$$

$$iv) \frac{(0.345+0.255)^2 - (0.345-0.255)^2}{0.345 \times 1.02}$$

Solution: given expression

$$= \frac{(0.345+0.255)^2 - (0.345-0.255)^2}{0.345 \times 1.02}$$

$$= \frac{4 \times 0.345 \times 0.255}{4 \times 0.345 \times 1.02}$$

$$[\because (a+b)^2 - (a-b)^2 = 4ab]$$

$$= 1$$

$$(v) \quad \frac{0.682 \times 0.682 - 0.318 \times 0.318}{0.682 - 0.318}$$

Solution : Given expression

$$= \frac{(0.682)^2 - (0.318)^2}{0.682 - 0.318}$$

$$= (0.682 + 0.318)$$

$$\left[\because \frac{a^2 - b^2}{a - b} = a + b \right]$$

$$= 1$$

$$(vi) \quad \frac{(3.29)^2 - (0.81)^2}{4}$$

Solution : given expression

$$= \frac{(3.29)^2 - (0.81)^2}{3.29 + 0.81}$$

$$= (3.29 - 0.81)$$

$$\left[\because \frac{a^2 - b^2}{a + b} = a - b \right]$$

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$$= 2.48$$

$$(vii) (2.35)^3 + 1.95 \times (2.35)^2 + 7.05 \times (0.65)^2 + (0.65)^3$$

Solution: Given expression

$$= (2.35)^3 + 3 \times 0.65 \times (2.35)^2$$

$$+ 3 \times 2.35 \times (0.65)^2 + (0.65)^3$$

$$= (2.35 + 0.65)^3$$

$$[\because a^3 + 3a^2b + 3ab^2 + b^3 = (a + b)^3]$$

$$= (3)^3$$

$$= 27.$$

$$(viii) \frac{(4.23)^3 - 3 \times 0.32 \times (4.32)^2 + 3 \times 4.32 \times (0.32)^2 - (0.32)^3}{4 \times 4 \times 4}$$

Solution: given expression

$$= \frac{(4.23)^3 - 3 \times 0.32 \times (4.32)^2 + 3 \times 4.32 \times (0.32)^2 - (0.32)^3}{4 \times 4 \times 4}$$

$$\frac{(4.32 - 0.32)^3}{4^3}$$

$$[\because a^3 - 3a^2b + 3ab^2 - b^3 = (a - b)^3]$$

$$= \left(\frac{4}{4}\right)^3$$

$$= 1$$

$$(ix) \frac{885 \times 885 \times 885 + 115 \times 115 \times 115}{885 \times 885 + 115 \times 115 - 885 \times 115}$$

$$\text{Solution: } = \frac{(885)^3 + (115)^3}{(885)^2 + (115)^2 - 885 \times 115}$$

$$(885 + 115)$$

$$[\because \frac{a^3 + b^3}{a^2 - ab + b^2} = a + b]$$

$$= 1000$$

$$(x) \frac{0.62 \times 0.62 \times 0.62 - 0.41 \times 0.41 \times 0.41}{0.62 \times 0.62 + 0.62 \times 0.41 + 0.41 \times 0.41}$$

Solution: given expression

$$= \frac{(0.62)^3 - (0.41)^3}{(0.62)^2 + 0.62 \times 0.41 + (0.41)^2}$$

$$= (0.62 - 0.41)$$

$$[\because \frac{a^3 - b^3}{a^2 + ab + b^2} = a - b]$$

$$= 0.21$$

$$(xi) \frac{(2.3)^3 + (1.5)^3 + (1.2)^3 - 3 \times 2.3 \times 1.5 \times 1.2}{2.3 \times 2.3 + 1.5 \times 1.5 + 1.2 \times 1.2 - 2.3 \times 1.5 - 2.3 \times 1.2 - 1.5 \times 1.2}$$

Solution: Given expression

$$\frac{(2.3)^3 + (1.5)^3 + (1.2)^3 - 3 \times 2.3 \times 1.5 \times 1.2}{(2.3)^2 + (1.5)^2 + (1.2)^2 - 2.3 \times 1.5 - 2.3 \times 1.2 - 1.5 \times 1.2}$$

$$= (2.3 + 1.5 + 1.2)$$

$$[\because \frac{a^3 + b^3 + c^3 - 3abc}{a^2 + b^2 + c^2 - ab - ac - bc} = a + b + c]$$

$$= 5$$

surds and indices are called, 'a surd' if n is a fraction and is called, 'an index' if n is an integer. a is called, 'the base'.

Common Errors While Using BODMAS Rule

Someone can make some common mistakes while applying the BODMAS rule to solve an expression and those mistakes are given below,

1. The presence of multiple brackets may cause confusion and we may get the wrong answer.

Analytical Skills-I

2. Error occurs due to lack of proper understanding of addition and subtraction of integers.

For example: $2-5+6=-3+6=3$.

But if we simplify $2-5+6=2-11=-9$, we will get a wrong answer.

3. By assuming that the division has higher precedence over multiplication and addition has higher precedence than the subtraction.

Multiplication and division are same level operations and must be performed from left to right sequence (whichever comes first in the expression) and same with addition and subtraction which are same levels operations to be performed after multiplication and division. If one solves division first before multiplication (which is on the left side of division operation) as D comes before M in BODMAS, they may get the wrong answer.

Summary

The key concepts learned from this unit are: -

- Understand what BODMAS Rule is.
- Understand conversion of symbols into signs.

Keywords

- BODMAS Rule.
- Precedence
- Symbols into signs.

Self Assessment

1. $3+(6+7)=$
A. 16
B. 17
C. 18
D. 12
2. $15-(3+2)=$
A. 16
B. 17
C. 18
D. 10
3. $2 \times (3 \times 8)=$
A. 16
B. 22
C. 18
D. 10
4. $[(50-(2+3))+15] \div 12=$

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- A. 5
B. 1
C. 10
D. 20
5. $8+9\div 9+5\times 2-7=$
A. 12
B. 15
C. 20
D. 25
6. $[25-3(6+1)]\div 4+9=$
A. 10
B. 12
C. 15
D. 19
7. $180\div 15\{(12-6)-(14-12)\}=$
A. (a)12
B. (b)6
C. (c)48
D. (d)20
8. $3+2^4\times(15\div 3)$ using BODMAS rule is
A. 24
B. 20
C. 25
D. 83
9. If '+' means 'divided by', '-' means 'multiplied by', 'x' means 'minus' and '□' means 'plus', which of the following will be the value of the expression $16\div 8-4+2\times 4$?
A. 16
B. 28
C. 32
D. 44
10. If A means 'plus', B means 'minus', C means 'divided by' and D means 'multiplied by', then $18\text{ A }12\text{ C }6\text{ D }2\text{ B }5=?$
A. 15
B. 25
C. 27
D. 17
11. If a means 'plus', b means 'minus', c means 'multiplied by', d means 'divided by', then $18\text{ c }14\text{ a }6\text{ b }16\text{ d }4=?$

Analytical Skills-I

- A. 63
- B. 254
- C. 288
- D. 1208

12. If x stands for 'add', y stands for 'subtract', z stands for 'divide' and p stands for 'multiply', then $(7 \div 3) \times y$
 6×5 ?

- A. 5
- B. 10
- C. 15
- D. 20

13. If A stands for +, B stands for -, C stands for $\times$, then $(10 \div 4) \times A (4 \div 4) \times B 6$?

- A. 60
- B. 56
- C. 50
- D. 46

14. Which of the following two signs need to be interchanged to make the given equation correct?

$$10 + 10 \div 10 - 10 \times 10 = 10$$

- A. + and -
- B. + and $\div$
- C. + and $\times$
- D. $\div$ and +

15. If A means -, B means $\div$, C means + and D means $\times$, then $15 \div 3 \times 24 \times 12 \div 2 = ?$

- A. 34
- B. 2
- C. $(\frac{9}{5})$
- D. none of these

Answers for Self Assessment

- | | | | | |
|-------|-------|-------|-------|-------|
| 1. A | 2. D | 3. B | 4. A | 5. A |
| 6. A | 7. C | 8. D | 9. B | 10. D |
| 11. B | 12. D | 13. C | 14. C | 15. A |

Review Questions

- 1. $15 + (6 + 7) =$
- 2. $25 - (3 + 2) =$
- 3. $2 \times (3 \times 8) =$

4. $[(10-(2+3))+15]\div 12=$
5. $18+19\div 19+15\times 12-17=$
6. $180\div 15\{(12-6)-(14-12)\}=$
7. Using BODMAS rule solve $30+25\times(25\div 30)$
8. If '-' means 'divided by', '+' means 'multiplied by', 'x' means 'minus' and ' $\div$ ' means 'plus', which of the following will be the value of the expression $16\div 8-4+2\times 4$?
9. If A means 'plus', B means 'divided by', C means 'minus' and D means 'multiplied by', then $18\ A\ 12\ C\ 6\ D\ 2\ B\ 5=$?
10. If a means 'plus', b means 'minus', c means 'multiplied by', d means 'divided by', then $18\ c\ 14\ a\ 6\ b\ 16\ d\ 4=$?
11. If x stands for 'divide', y stands for 'subtract', z stands for 'add' and p stands for 'multiply', then $(7\ p\ 3)\ y\ 6\ x\ 5$?



Further Readings

1. Quantitative Aptitude For Competitive Examinations By Dr. R S Aggarwal, S Chand Publishing
2. A Modern Approach To Verbal & Non-Verbal Reasoning By Dr. R S Aggarwal, S Chand Publishing
3. Magical Book On Quicker Maths By M Tyra, Banking Service Chronicle
4. Analytical Reasoning By M.K. Pandey, Banking Service Chronicle